

Generation Interconnection Request System Impact Study Report

For Queue Project Q24-2390

Pines 100 kV
March 16, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2390. The Interconnection Customer proposes a 118.4 MW solar photovoltaic facility interconnecting at the Pines 100 kV substation in the Ohio area. The request was received on March 07, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 503 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,323,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Pines – Patrick Substation 100.0 kV
Contingency: East Marysville – Madeline 100.0 kV
Loading: 124.8% (Rating: 128 MVA)
DFAX: 48.9%
Pre-project loading: 79.6%
2. Facility: Hillsborough County Resource Recovery – Lawther Dr 100.0 kV
Contingency: Baraga Vlg 500/230kV TX
Loading: 120.6% (Rating: 125 MVA)
DFAX: 5.0%
Pre-project loading: 115.9%
3. Facility: Roe'S Corner – Thornydale 100.0 kV
Contingency: Braga Vlg 500-230 kV TX
Loading: 120.0% (Rating: 115 MVA)
DFAX: 17.1%
Pre-project loading: 102.4%
4. Facility: Patick – Gridley 100.0 kV
Contingency: Baraga Vlg 500-230 kV XFMR
Loading: 111.2% (Rating: 122 MVA)
DFAX: 18.0%
Pre-project loading: 93.7%
5. Facility: Miami – Fitchburg 100.0 kV L
Contingency: Miami – Fiddymment 100.0 kV
Loading: 108.9% (Rating: 163 MVA)
DFAX: 9.3%
Pre-project loading: 102.1%

6. Facility: Otter Vly – Blaney Park Sw Sta 100.0 kV Cir. 1
Contingency: Otter Vly – Blaney Park Sw Sta 100.0 kV Line 2
Loading: 107.7% (Rating: 151 MVA)
DFAX: 5.1%
Pre-project loading: 103.7%
7. Facility: Blaney Park Sw Sta – Otter Valley Substation 100.0 kV L
Contingency: Otter Vly – Blaney Park Sw Sta 100.0 kV Cir. 1
Loading: 107.7% (Rating: 151 MVA)
DFAX: 5.1%
Pre-project loading: 103.7%
8. Facility: Jones Canyon – Thornydale 100.0 kV L
Contingency: Baraga Vlg 500/230kV XFMR
Loading: 106.5% (Rating: 121 MVA)
DFAX: 17.1%
Pre-project loading: 89.8%
9. Facility: Patrick – Roe'S Corner 100.0 kV line
Contingency: Baraga Vlg Substation 500/230kV XFMR
Loading: 106.0% (Rating: 142 MVA)
DFAX: 17.1%
Pre-project loading: 91.8%
10. Facility: Fitchburg – W Riverside 100.0 kV L
Contingency: Fiddymment – Miami 100.0 kV
Loading: 104.9% (Rating: 197 MVA)
DFAX: 9.3%
Pre-project loading: 99.3%
11. Facility: Lons Road – Aes Beaver Valley Partners Beaver Vly 500.0 kV
Contingency: Cranberry – Beaver Vly 500.0 kV
Loading: 103.1% (Rating: 2,652 MVA)
DFAX: 7.9%
Pre-project loading: 102.8%
12. Facility: Cranberry – Beaver Vly 500.0 kV
Contingency: Aes Beaver Valley Partners Beaver Vly – Lions Road 500.0 kV line
Loading: 102.7% (Rating: 2,657 MVA)
DFAX: 7.9%
Pre-project loading: 102.3%
13. Facility: N Msquite Tap – Hillsborough County Resource Recovery 100.0 kV
Contingency: Baraga Vlg 500 kV/230 kV TX
Loading: 102.4% (Rating: 131 MVA)
DFAX: 5.0%
Pre-project loading: 97.9%
14. Facility: Caro – Miami 100.0 kV Line
Contingency: Baraga Vlg 500/230 kV Transformer Bank
Loading: 100.0% (Rating: 310 MVA)
DFAX: 7.4%
Pre-project loading: 97.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pines – Patrick Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$5,033,000	100% (\$5,033,000)
Lawther Dr – Hillsborough County Resource Recovery 100.0 kV L	Rebuild line with higher-capacity conductor	\$6,869,000	100% (\$6,869,000)
Roe'S Corner – Thornydale 100.0 kV	Rebuild line with higher-capacity conductor	\$8,953,000	100% (\$8,953,000)
Patrick – Gridley 100.0 kV	Rebuild line with higher-capacity conductor	\$3,966,000	100% (\$3,966,000)
Fitchburg – Miami 100.0 kV	Reconductor line	\$659,000	38% (\$249,000)
Blaney Park Sw Sta – Otter Valley 100.0 kV	Reconductor line	\$320,000	100% (\$320,000)
Blaney Park Sw Sta – Otter Valley Substation 100.0 kV L	Reconductor line	\$410,000	100% (\$410,000)
Jones Canyon – Thornydale 100.0 kV L	Reconductor line	\$2,092,000	100% (\$2,092,000)
Roe'S Corner – Patrick 100.0 kV	Reconductor line	\$894,000	100% (\$894,000)
Fitchburg – W Riverside 100.0 kV	Reconductor line	\$1,494,000	38% (\$564,000)
Lions Road – Aes Beaver Vly Partners Beaver Vly 500.0 kV	Reconductor line	\$32,549,000	1% (\$290,000)
Cranberry – Beaver Vly 500.0 kV	Reconductor line	\$31,817,000	1% (\$283,000)
N Msquite Tap – Hillsborough County Resource Recovery 100.0 kV	Reconductor line	\$835,000	100% (\$835,000)
Caro – Miami 100.0 kV Line	Reconductor line	\$1,597,000	35% (\$565,000)
	Total allocated to project		\$31,323,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.